

# AI-Driven Software Design and Applications: Construction of Personalized Learning Measurement Models Under the Taxonomic Development Perspective

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**Abstract**—To address the demand for differentiated learning resources, this study designed and developed an AI-driven software using the PyCharm platform. The research framework consisted of six key stages. First, a composite function was designed to evaluate students' academic foundations and learning capabilities, integrating historical course performance, classroom engagement, and teacher assessments. Second, the Qwen AI model was pre-trained on the Alibaba Cloud platform using four sets of domain-specific datasets. Third, a PC-based educational software was developed using PyCharm, with API integration enabling seamless communication between the local system and the Qwen AI model. Fourth, Student Baseline Data (SBD) were input into the software as feedforward information. Fifth, the API communication protocol was optimized to guide the AI model in incorporating feedforward data, thereby generating personalized responses for different user profiles. Finally, the software was deployed in a real classroom setting to collect user feedback. A controlled experiment was conducted with a control group (n=20) and an experimental group (n=17). Evaluation results showed that the experimental group achieved a mean score improvement of 0.0452 (experimental: 0.8847 vs control: 0.8395), with variance ratios further optimized from 0.0025 to 0.0039. Statistical significance was confirmed by t-test analysis ( $t=-2.4622$ ,  $p=0.0189$ ). The software demonstrated fast response times, accurate content delivery, and highly personalized tutoring capabilities. This study highlights the transformative potential of AI in education, particularly in delivering customized learning experiences, and contributes a practical framework for integrating AI into vocational pedagogy.

**Keywords**—software design; Qwen model; AI; AI-driven; t-test

## I. INTRODUCTION

In traditional educational settings, the uniform approach often falls short in addressing individual developmental needs. Vocational students, in particular, show considerable variation in prior knowledge and learning capacities. Yet, instructional strategies and resources typically remain static, offering little flexibility to accommodate diverse learner requirements. This rigidity poses a significant barrier to effective education, limiting

advanced learners' opportunities for challenge and struggling students' access to necessary support. Traditional pedagogy's inability to adapt exacerbates these issues [1, 2]. The Taxonomic Development Perspective (TDP) offers a solution by emphasizing personalized learning pathways, especially vital in subjects like material mechanics that span from basic principles to sophisticated applications. Our research adopts material mechanics as a case study, developing PC-based software using PyCharm and integrating Qwen's API to deliver customized content. This initiative not only tackles educational equity but also showcases AI's potential in transforming teaching practices, making it a pivotal contribution to both educational technology and applied AI domains [3, 4].

## II. LITERATURE REVIEW

In recent decades, education scholars have extensively investigated the integration of AI into teaching practices, aiming to enhance learning efficiency and adaptability. Y. A. Alshumaimeri [5] conducted a systematic review highlighting AI's role in optimizing learning path planning for foreign language education. J. Álvarez Ariza et al. [6] expanded on this by analyzing 146 studies, revealing that generative AI significantly contributes to K-12 and higher education through automated content generation, adaptive assessments, and advanced learning analytics. A. F. M. Fahimey [7] explored AI's impact on Learning Management Systems (LMS) within Malaysian vocational education. R. A. Abdel-Rahem [8] emphasized the need for strategic AI integration and instructor training at the University of Petra to prepare for future educational transformations. T. Akila [9] further illustrated how AI-powered platforms such as Chatbot. S. Palarimath [10] integrated AI with multiple intelligences to develop an adaptive educational model that fosters creativity and problem-solving skills, while V. K. Stephen [11] introduced an AI-based teaching system using the IFH-DaBoost algorithm. C. C. Orlanda-Ventayen [12] also stressed the importance of ethical frameworks to ensure responsible AI implementation in education. Parallel to AI advancements, personalized learning has emerged as a key focus in educational research, driven by technologies that

cater to individual learner needs. Y. Han [13] proposed a multidimensional course recommendation method based on a two-layer Hidden Markov Model (TH3MPCR), which improved personalization accuracy and interpretability in online education. L. Lina [14] developed a user-centric e-learning system with a multi-layer architecture that captures individual learning characteristics, enhancing personalization through intelligent information filtration. L. Min [15] introduced a learning resource recommendation algorithm combining knowledge graphs with interest diffusion mechanisms. X. Pan [16] designed a deep learning-based multi-view course recommendation system that extracts key features from student and course models to generate customized suggestions. N. Wan [17] further

advanced personalized learning by constructing a big data-driven framework. These studies collectively demonstrate a growing shift toward highly adaptive, data-driven educational environments that prioritize individualized learning pathways.

### III. METHODOLOGY

#### A. Research Logical Framework

The study's research logic, as illustrated in Fig. 1, is structured into three interrelated phases: theoretical foundation, system architecture, and empirical evaluation, forming a cyclical feedback loop to iteratively refine the AI-driven educational framework.

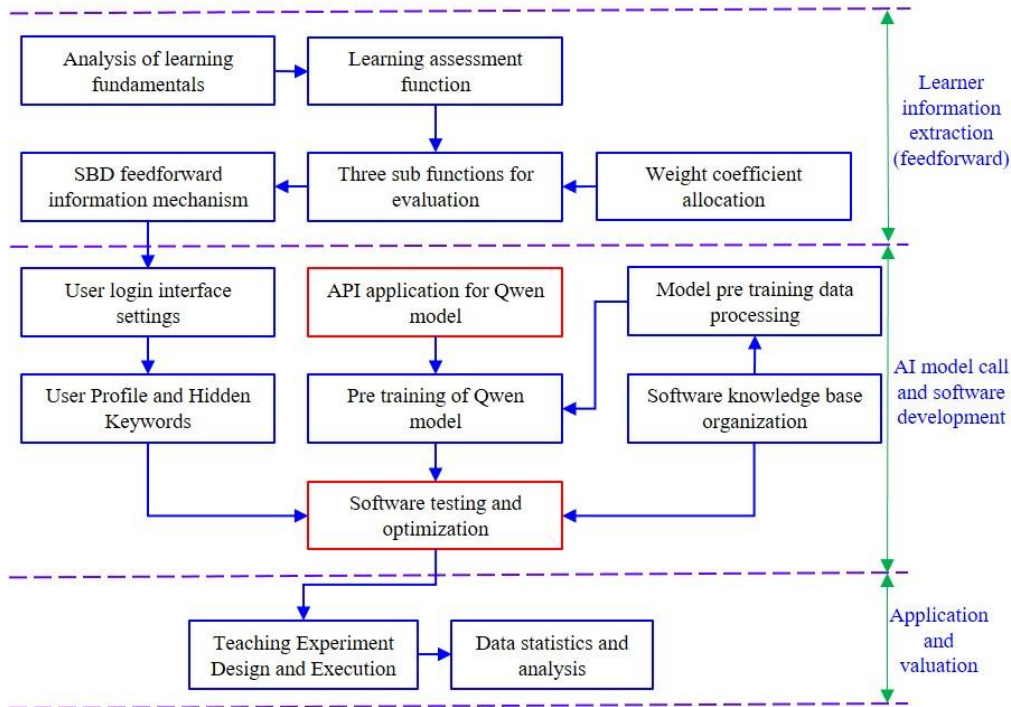


Fig. 1. The research logic diagram of this study.

The theoretical phase establishes a learning assessment function, decomposed into knowledge acquisition tracking, skill proficiency estimation, and adaptive feedback generation, with dynamic weight coefficients calibrated based on learner profiles. This guides the design of the SBD feedforward mechanism, which prioritizes personalized content delivery by extracting contextual metadata. The system architecture phase integrates modular components, including a user login interface, dynamic profile database, and SBD feedforward module, bridging theoretical models with technical execution. Technical implementation involves pre-training the Qwen model on domain-specific datasets and developing PC-based software via PyCharm, ensuring API integration for real-time interaction. Finally, empirical evaluation validates the system through controlled classroom experiments, measuring performance improvements and feedback efficacy. Data-driven analysis,

including t-tests and variance comparisons, confirms the framework's effectiveness in enhancing learning outcomes.

#### B. The Gantt Chart of this Study

The research framework, illustrated in Fig. 2, outlines a structured five-part methodology aligned with the TDP to construct AI-driven personalized learning models for vocational education. Task-1 (overall framework design) initiates the process, sequentially addressing design (Week 0–1), processing (Week 1–2), and optimization (Week 2–12), establishing the foundational architecture for subsequent stages. Task-2 (student baseline data) follows, with design (Week 1–3), processing (Week 3–5), and optimization (Week 5–7), ensuring tailored stratification of learners' academic profiles. Concurrently, Task-3 (AI model pre-training) spans design (Week 2–4) and processing (Week 4–9), integrating domain-specific datasets to enhance Qwen's precision in adaptive responses.

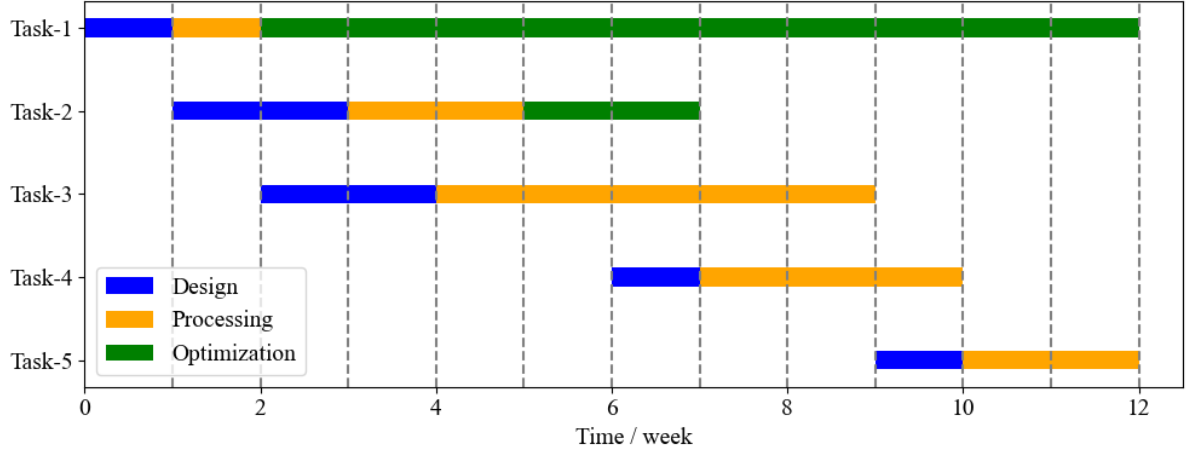


Fig. 2. The Gantt chart of this study.

Task-4 (software development) then focuses on design (Week 6–7) and processing (Week 7–10), leveraging PyCharm to implement taxonomic functionalities. Finally, Task-5 (pedagogical experiments and data analysis) concludes the framework, with design (Week 9–10) and processing (Week 10–12), validating the system’s efficacy through empirical evaluation.

### C. Student Baseline Data

The SBD are characterized by the matrix shown in (1).

$$\mathbf{SBD} = \begin{bmatrix} Y_1(x) \\ Y_2(x) \\ Y_3(x) \end{bmatrix} \quad (1)$$

Here, the function  $Y_1(x)$  represents the academic foundation of student  $x$ , expressed as (2).

$$Y_1(x) = T_1(x) + S_1(x) \quad (2)$$

The function  $T_1(x)$  denotes the teacher’s evaluation of student  $x$ ’s academic foundation. The function  $S_1(x)$  is the equivalent score derived from the previous  $n$  course examinations of student  $x$ , formulated as (3):

$$S_1(x) = \frac{\alpha \sum_{i=1}^n \text{score}(i)}{\sum_{i=1}^n \text{Max\_score}(i)} \quad (3)$$

Where  $\text{score}(i)$  is the examination score for course  $i$ ,  $\text{Max\_score}(i)$  is the maximum score for course  $i$ , and  $\alpha$  is the weighting coefficient. The function  $Y_2(x)$  quantifies student  $x$ ’s learning engagement, defined in (4).

$$Y_2(x) = T_2(x) + S_2(x) \quad (4)$$

$T_2(x)$  reflects the teacher’s assessment of student  $x$ ’s engagement level.  $S_2(x)$  represents the engagement metrics from the previous  $n$  courses, calculated via (5), where

$\text{partic}(i)$  is the engagement score for course  $i$  (extracted from the Learning Companion APP), and  $\text{Max\_partic}(i)$  is the maximum engagement score for course  $i$  (also from the APP). The coefficient  $\beta$  serves as the weighting factor.

$$S_2(x) = \frac{\beta \sum_{i=1}^n \text{partic}(i)}{\sum_{i=1}^n \text{Max\_partic}(i)} \quad (5)$$

Finally,  $Y_3(x)$  as shown in (6), characterizes student  $x$ ’s learning capacity, determined by performance on challenging questions in prior course exams, with  $\lambda$  as the corresponding weighting coefficient.

$$Y_3(x) = \frac{\lambda \sum_{i=1}^n \text{diffi}(i)}{\sum_{i=1}^n \text{Max\_diffi}(i)} \quad (6)$$

These stratified metrics enable the system to categorize learners into taxonomic tiers—foundational, intermediate, and advanced—thereby aligning AI-driven content delivery with individual readiness levels. The temporal integration of historical performance data ( $n$  courses) further enhances predictive accuracy, creating a dynamic baseline that adapts to evolving learner profiles throughout the pedagogical intervention.

The parameter configuration of the proposed framework was determined through collaborative discussions with six experienced vocational educators, ensuring alignment with pedagogical principles and practical feasibility. The main parameters of the algorithm are shown in Tab. 1. For the variable  $n$ , representing the number of prerequisite courses used to assess students’ academic foundations, teachers voted on its value. The mean score was 3.2, rounded to 3 for operational simplicity, indicating that only the best three course performances are considered in evaluating a student’s pre-learning status for

the material mechanics course. Regarding the teacher-assigned scores  $T_1$  (academic foundation) and  $T_2$  (learning engagement), the average maximum scores suggested by educators were 9.6/100 and 13.6/100, respectively. To simplify computational complexity and ensure balanced weighting, both were standardized to  $T_1=T_2=10$ , meaning teacher evaluations account for 10% of the total weight in the baseline data model. This decision emphasizes objective data (e.g., learning companion app metrics) over subjective teacher judgments, with the weighting coefficients  $\alpha$  and  $\beta$  each set to 90, reflecting their dominant 90% contribution to the final assessment. For the coefficient lambda, which governs the weight of high-difficulty exam scores in measuring learning capacity ( $Y_3(x)$ ), teachers' average rating was 156.4. To maintain numerical simplicity,  $\lambda=150$  was adopted, underscoring the consensus that performance on challenging questions is critical for evaluating learners' problem-solving abilities. This parameterization ensures the model prioritizes empirical data while retaining interpretability, enabling the

AI system to deliver educationally grounded, data-driven interventions.

TABLE I. SETTING OF MAIN PARAMETERS

| <i>The numerical values or ranges of the main parameters</i> |         |           |
|--------------------------------------------------------------|---------|-----------|
| $n$                                                          | $T_1$   | $T_2$     |
| 3                                                            | 0-10    | 0-10      |
| $\alpha$                                                     | $\beta$ | $\lambda$ |
| 90                                                           | 90      | 150       |

#### D. Pre Training of AI models

Building on prior experience with Alibaba Cloud's AI models [11], this study leveraged the Qwen model for pre-training to enhance its efficiency and intelligence in delivering personalized educational services. The pre-training process was implemented via the Alibaba Cloud platform, as illustrated in Fig. 3, where the "Add Dataset" button (Fig. 4) enabled the upload of four XLSX-formatted databases.

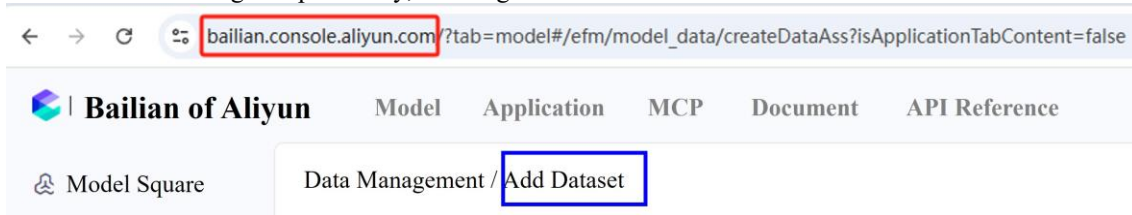


Fig. 3. The page (URL) for pre training execution.

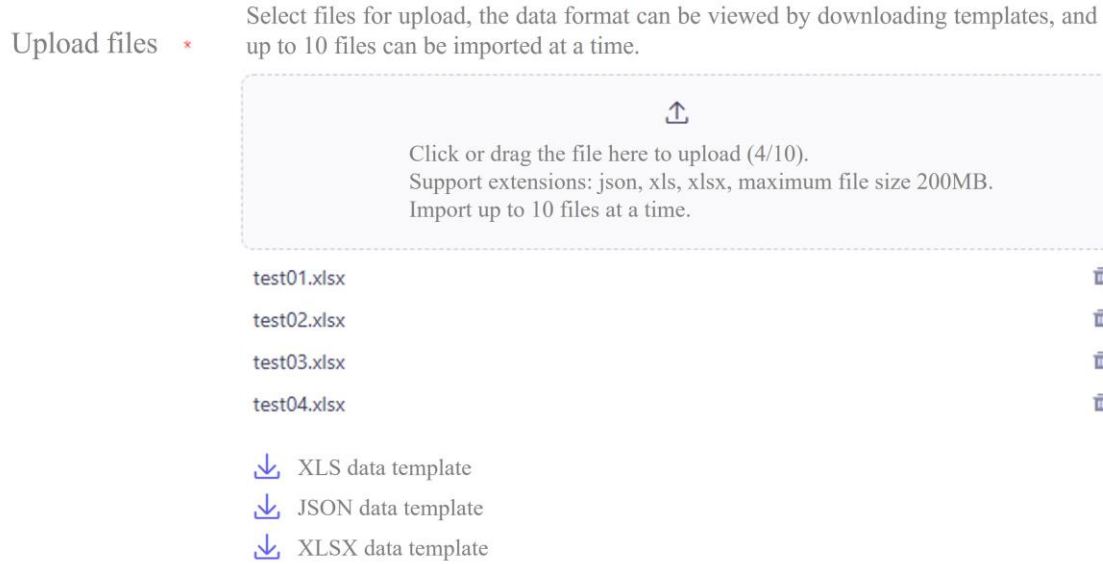


Fig. 4. Submission of pre training database (screenshot of Alibaba Cloud backend).

These datasets, containing structured academic records, learning behavior metrics, and domain-specific material mechanics content, were curated to align with the TDP. During pre-training, the model underwent parameter optimization to prioritize task-specific knowledge extraction, such as differentiating between foundational and advanced learners' needs. For instance, the model learned to contextualize responses to material mechanics

problems by integrating SBD (e.g., historical exam scores, engagement levels) from the uploaded datasets. This customization ensured that post-training, the Qwen model could generate stratified explanations—simplified for beginners and analytically rigorous for advanced learners—while maintaining technical accuracy. Post-pre-training validation confirmed improved alignment with pedagogical objectives, as evidenced by the model's ability



to dynamically adapt its output complexity based on user-level metadata.

#### E. API Calls for AI Models

The local PC software communicates with the Qwen AI model through an API interface, as shown in the code screenshot (Fig. 5). The API endpoint is provided by Alibaba Cloud, ensuring stable and secure data exchange. To prevent unauthorized access, the API key used in the request has been masked in the figure.

```
# Alibaba Cloud Bailian API Configuration
API_URL = "https://dashscope.aliyuncs.com/api/v1/services/ai"
API_KEY = "skXXXXXXXX1391c1"

class ChatApplication:
    def __init__(self, master):
        self.master = master
        master.title("Qwen-turbo-API")
        master.geometry("1000x600")
        master.configure(bg="#f0f4f8")
```

The call of Qwen Model

Fig. 5. API calls for AI models (Code screenshot).

This integration enables users to input queries locally and receive real-time responses from the Qwen model hosted remotely. Unlike general web-based Q&A systems, this software leverages the API to incorporate user-specific features into the query process. For example, prior learning data and user profiles can be appended to each request, allowing the AI model to generate more accurate and personalized answers. This capability highlights the advantage of combining local software functionality with cloud-based AI services, offering a scalable solution for intelligent education systems.

#### F. Software Interface and Functions

The software assigns each student a unique ID linked to their SBD information, including  $F_1$ ,  $F_2$ , and  $F_3$  values that represent individual learning characteristics. As shown in the test version of the login interface (Fig. 6), entering a student ID displays the corresponding F-values. The system passes SBD values to the Qwen model before generating responses, enabling personalized feedback based on individual learning profiles.

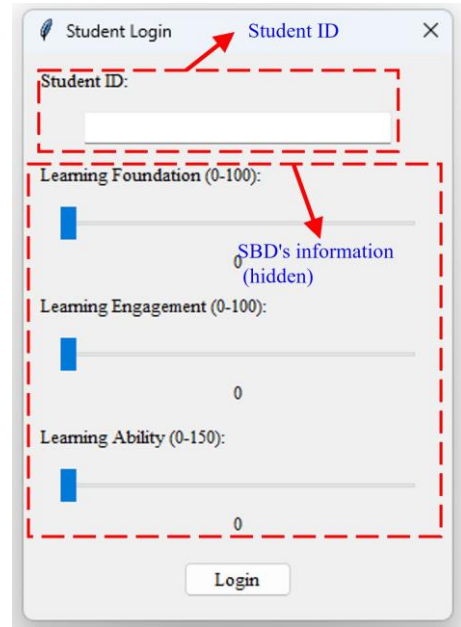


Fig. 6. User login interface (beta version, software screenshot).

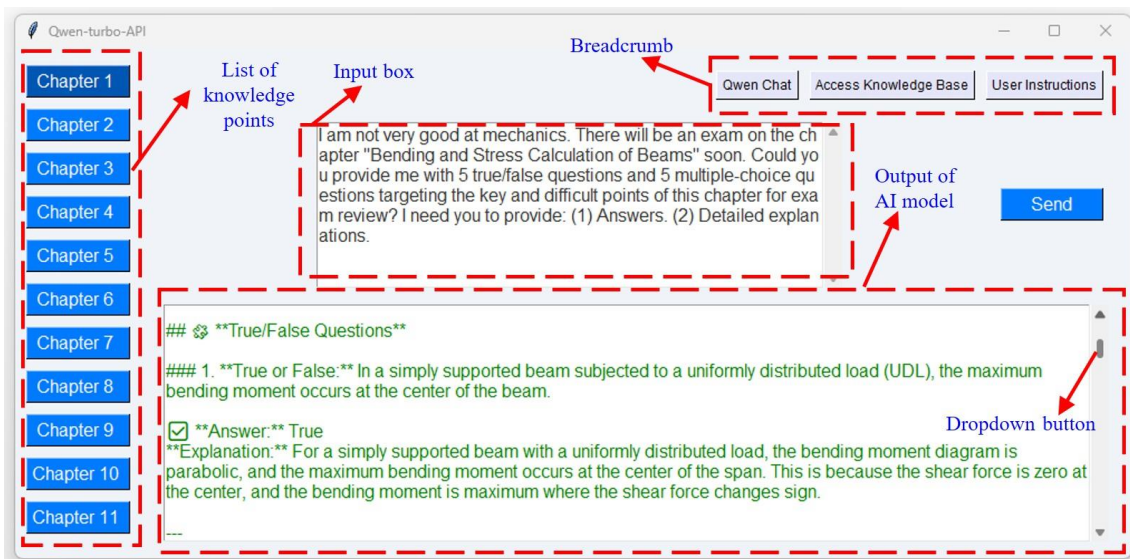


Fig. 7. The question and answer module of the learning software. (software screenshot).

In the final release, these fields will be hidden (red box in Fig. 6) to protect privacy while still supporting internal personalization. Addressing learner diversity is essential in vocational education, especially in subjects like mechanics of materials where concept complexity varies widely [6].

Differentiated instruction allows alignment of content difficulty with student readiness, improving both engagement and understanding. AI integration enables real-time adaptation of learning paths, ensuring efficient and effective support tailored to each learner. The software

interface (Fig. 7) is structured to enhance usability and learning efficiency. It includes two main modules.

The first is the “knowledge point navigation” panel on the left, which organizes course content by chapters, allowing students to quickly access and review targeted materials. This modular design supports self-paced, focused study. The second core module is the Qwen-based Q&A system, which serves as an intelligent tutoring component. Students can input their questions into a text box and click the “send” button (as shown in Fig. 7) to receive immediate responses generated by the Qwen AI model. This real-time interaction mechanism greatly reduces the time needed to resolve academic queries, enhancing overall learning efficiency. Natural language processing enables the system to understand and respond accurately to diverse questions. A breadcrumb-style navigation bar with a “user instructions” button is placed at the top right corner, offering guided support for new users. These features promote an interactive, accessible, and learner-centered environment. The integration of structured content delivery and AI-driven assistance aligns with differentiated instruction principles, demonstrating the system’s adaptability to varied learning needs in vocational education [6].

#### IV. RESULTS AND DISCUSSION

##### A. Data Analysis

This study conducted a controlled teaching experiment to evaluate the efficacy of the AI-driven software in enhancing learning outcomes. The control group ( $n=20$ ) followed traditional classroom instruction, while the experimental group ( $n=17$ ) used the developed software as a supplementary tool over a two-week period (4 class hours). Post-experimental evaluations were collected via a WeChat Mini Program, requiring students to submit two-digit scores (out of 100). To standardize data for analysis, scores were normalized by multiplying by 0.01, converting the maximum value to 1. The results are shown in Tab. 2. Statistical analysis revealed that the experimental group achieved a higher mean score (0.8847) compared to the control group (0.8395), with a 0.0452 absolute difference. Variance analysis further indicated reduced variability in the experimental group ( $\text{variance}=0.0025$ ) relative to the control group ( $\text{variance}=0.0039$ ), suggesting more consistent performance among users of the AI tool. An independent samples t-test confirmed the significance of this difference ( $t=-2.4622$ ,  $p=0.0189$ ,  $p<0.05$ ), rejecting the null hypothesis of no difference between groups.

TABLE II. TESTING RESULTS OF TWO GROUPS

| <i>Descriptive statistics and hypothesis testing results of two groups</i> |                             |                                  |                   |
|----------------------------------------------------------------------------|-----------------------------|----------------------------------|-------------------|
| <i>Metric</i>                                                              | <i>Control Group (n=20)</i> | <i>Experimental Group (n=17)</i> | <i>Test Value</i> |
| <i>Mean</i>                                                                | 0.8395                      | 0.8847                           | $t=-2.4622$       |
| <i>Variance</i>                                                            | 0.0039                      | 0.0025                           | $p=0.0189^*$      |

These results demonstrate that the software not only improved average performance but also enhanced learning stability, aligning with the TDP emphasis on stratified, personalized instruction. The reduction in variance implies

the tool’s capacity to address heterogeneous learner needs, minimizing disparities in academic outcomes. The statistical significance of the findings underscores the potential of AI-integrated educational frameworks to optimize pedagogical strategies in vocational training. Future work will explore longitudinal effects and scalability across larger cohorts.

##### B. Teacher-Student Discussion

To further validate the software’s effectiveness, a post-experiment discussion was held with five students from the experimental group. All participants praised its practicality and impact on learning, highlighting three main benefits: fast problem-solving, personalized support, and increased motivation. Student A noted the tool’s quick response time, which helped clarify difficult mechanics concepts instantly, improving focus and confidence. This supports the improved performance data ( $\text{mean}=0.8847$ ). Student B highlighted how the system adapted to their pace, identifying weak areas in stress analysis and offering targeted exercises—likely contributing to the lower variance (0.0025). Student C emphasized the motivational effect of the interactive interface, encouraging deeper engagement with the course material. This aligns with the 0.0452 score improvement over the control group. Overall, these insights confirm the tool’s role in bridging theory and practice while supporting individualized, data-driven instruction in vocational education.

##### C. Discussion

This study contributes to existing research on AI applications in education, particularly in personalized and vocational learning. The developed software aligns with findings showing that AI improves student engagement and comprehension through intelligent tutoring systems [6]. Like Abdel-Rahem et al., we found that AI integration enables timely, accurate, and individualized support [8]. Our work extends this by introducing a feedforward mechanism based on SBD, improving the relevance of AI-generated responses—supporting Fahimey et al.’s view that AI should respond to learners’ academic backgrounds [7]. The observed improvements in mean scores and reduced variance suggest AI’s role in standardizing outcomes across diverse learners, echoing Alshumaimeri’s findings in language learning [5]. The API-based communication structure resembles layered models described by Lina and Han et al., but our system uniquely combines domain-specific training with real-time profiling for more context-aware responses [13, 14]. This advances Min’s idea of integrating knowledge graphs with interest modeling, though our focus is on immediate feedback [15]. Building on Álvarez Ariza et al. and Stephen et al. [6, 11], our research highlights how domain-specific AI can be applied effectively in complex technical fields like materials mechanics. Future work should focus on integrating real-time analytics to refine user models continuously. This enhancement could lead to more adaptive and responsive tutoring experiences. Moreover, the research team will explore the integration of digital twin and simulation technologies to enhance the software’s

capabilities in teaching mechanics [18, 19]. These technologies can provide interactive, risk-free environments for students to experiment with complex mechanical systems, thereby improving conceptual understanding and engagement.

## V. CONCLUSIONS

This study successfully demonstrated the efficacy of an AI-driven educational software in enhancing personalized learning experiences. By integrating historical performance and real-time feedback through a sophisticated API communication protocol, the software significantly improved learning outcomes, as evidenced by a mean score increase of 0.0452 in the experimental group compared to the control group. The reduced variance highlighted more consistent academic performance among students. Statistical analysis confirmed the significance of these improvements ( $t=-2.4622$ ,  $p=0.0189$ ), underscoring the potential of AI-integrated tools to deliver tailored educational support effectively. This research provides a robust framework for future implementations of AI in vocational education, promoting enhanced engagement and mastery among diverse learner populations.

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